Problem 00000000561: a square-zero Koszul algebra excludes every finite cyclotomic denominator

lizaixi01

24 October 2026

Precise scope. The printed conjecture asks that the Hilbert series of the Yoneda algebra of an N-Koszul algebra be rational with denominator $\prod_i (1 - t^{d_i})$. The product has no printed index set. We use the ordinary rational-function meaning of a denominator: a *finite polynomial* product, with positive integer d_i, allowing repetitions. Under this reading the denominator clause is false already for $N = 2$. The example's series is itself rational. We do not rule out an infinite formal Euler product, or dispute rationality for this example. The full original English and Chinese text is in `SOURCE.md`.

Theorem 1. *There is a connected quadratic 2-Koszul $\mathbb{Q}$-algebra A, with augmentation $k = \mathbb{Q}$, such that its actual graded Yoneda spaces satisfy*

$$\dim_{\mathbb{Q}} \operatorname{Ext}_A^n(k,k) = 2^n \quad (n \geq 0), \qquad H(t) = \sum_{n \geq 0} \dim_{\mathbb{Q}} \operatorname{Ext}_A^n(k,k) t^n = \frac{1}{1-2t}.$$

For every finite family of positive integers (d_i) and every $P \in \mathbb{Q}[t]$,

$$\left(\prod_i (1 - t^{d_i}) \right) H(t) \neq P(t).$$

The empty family is allowed as well.

1. The algebra and its full linear minimal resolution

Use the native square-zero extension

$$A = \mathbb{Q} \oplus V, \quad V = \mathbb{Q}^{\{0,1\}}, \qquad (r,v)(s,w) = (rs, rw + sv).$$

Let $\epsilon_a = (0, e_a)$ for the two genuine coordinate basis vectors e_a. All four products $\epsilon_a \epsilon_b$ vanish. The grading is $A_0 = \mathbb{Q} \oplus 0$, $A_1 = 0 \oplus V$, and $A_j = 0$ for $j \geq 2$. The formalization constructs its actual submodules, native internal direct-sum decomposition and native `GradedAlgebra` structure. The degree-zero component is linearly isomorphic to $\mathbb{Q}$ and the two ϵ_a span degree one.

This is a genuine quadratic presentation, rather than an assigned label:

$$A \simeq \mathbb{Q}\langle x_0, x_1 \rangle / (x_a x_b \mid a, b \in \{0,1\}).$$

Lean constructs the free algebra, the relation identifying each displayed length-two monomial with zero, and its native `RingQuot`. The free evaluation $x_a \mapsto \epsilon_a$ descends. In the reverse direction, a vector $v \in V$ maps to $\sum_a v_a \bar{x}_a$; products of any two such images vanish. The native square-zero-extension lift supplies the inverse homomorphism. The two composites are proved to be identities on all elements.

Let B_0 be the one-element set and $B_{n+1} = \{0,1\} \times B_n$. These are actual finite binary words, with cardinality proved inductively to be 2^n. Put $P_n = A^{B_n}$ and define actual A-linear maps

$$d_n : P_{n+1} \longrightarrow P_n, \qquad (d_n f)(w) = \sum_{a=0}^{1} \epsilon_a f(a,w), \qquad \eta : P_0 \longrightarrow k, \quad \eta(f) = f(*)_0.$$

Here $(r, v)_0 = r$. Coordinate calculations give, for every n, f, w, a,

$$(d_n f)(w)_0 = 0, \qquad (d_n f)(w)_1(a) = f(a, w)_0.$$

Consequently $d_n d_{n+1} = 0$, $\ker d_n$ consists exactly of functions with zero scalar coordinates, and $\operatorname{im} d_{n+1}$ is exactly that same submodule. For surjectivity onto this submodule, lift g by $f(a, w) = (g(w)_1(a), 0)$. The same calculation proves $\ker \eta = \operatorname{im} d_0$ and η is surjective. Thus the entire infinite complex is exact in every required degree.

Each P_n has the native finite free coordinate basis and is projective. Lean packages this exact augmented complex as an actual native `ProjectiveResolution`, including its quasi-isomorphism to $k[0]$. Give its basis vectors degree n: explicitly $(P_n)_j = A_{j-n}^{B_n}$ for $j \geq n$, and zero otherwise. The native component decompositions and graded scalar actions are constructed; every d_n preserves internal degree, and the augmentation preserves the grading of k concentrated in degree zero. On basis vectors,

$$d_n(e_{(a,w)}) = \epsilon_a e_w.$$

All matrix entries lie in degree one and in the actual augmentation ideal. The image is proved to equal the A-span of these positive-degree basis multiples. This is a minimal linear graded free resolution. It proves the ordinary Koszul defining criterion [1, Definition 8.1]; in the N-Koszul notation $\nu_N(i) = N\lfloor i/2 \rfloor + (i \bmod 2)$, Lean proves $\nu_2(i) = i$ for every i. This is the $N = 2$ case of the standard convention in [2, p. 4, Definition 2.2]. No theorem about Hilbert series is assumed.

2. Native Ext and the denominator obstruction

For every actual A-linear map $P_n \to k$, precomposition with d_n is zero, because every ϵ_a acts as zero on k. Evaluation on the genuine free basis gives a proved $\mathbb{Q}$-linear equivalence

$$\operatorname{Hom}_A(P_n, k) \simeq_{\mathbb{Q}} \mathbb{Q}^{B_n}.$$

The inverse sends a coordinate family c to $f \mapsto \sum_w f(w)_0 c_w$. Reconstruction is proved on every f. The native projective-resolution comparison constructs all elements of the native $\operatorname{Ext}_A^n(k, k)$ from such cocycles. Its actual surjectivity theorem and coboundary criterion show that the map is bijective for $n > 0$, since every possible boundary map is zero. Compatibility with the native $\mathbb{Q}$-scalar action is proved using Yoneda postcomposition. For $n = 0$, native $\operatorname{Ext}^0 \simeq \operatorname{Hom}_A(k, k)$ followed by evaluation at 1 gives the actual equivalence with $\mathbb{Q}$.

All native Ext spaces are therefore proved finite dimensional, and their finranks are derived from $|B_n| = 2^n$. The Hilbert series is the actual formal power series whose coefficient is this native finrank. In particular, no infinite-dimensional finrank default is used. The proof uses the graded native Ext spaces; it includes native bilinear Yoneda composition but does not construct a separate bundled total Yoneda ring. Multiplication is not needed for this Hilbert-series assertion.

The coefficient at zero is 1 and successive coefficients double. Lean therefore proves the formal-series identity

$$(1 - 2t)H(t) = 1.$$

Suppose a finite positive family gave $D(t)H(t) = P(t)$, where $D(t) = \prod_i (1 - t^{d_i})$. Multiplying by $1 - 2t$, commuting factors and using this identity gives $(1 - 2t)P(t) = D(t)$ as an identity of actual formal series. Injectivity of the native polynomial-to-power-series map makes this an identity of polynomials. Evaluating at $t = 1/2$ gives

$$0 = D(1/2) = \prod_i (1 - 2^{-d_i}) \neq 0,$$

because each $d_i > 0$. Lean proves this nonvanishing for every finite index type, including repeated exponents and the empty product. This contradicts the finite-denominator clause independently of the source's particular degree-sequence choice.

3. Formalization boundary and reproduction

All algebra, grading, full-degree exactness, projectivity, actual native Ext comparisons, dimensions, formal-series operations and the finite-denominator contradiction above are proved in `lean/Main.lean`. The conventional name "2-Koszul" is identified by its defining minimal-linear-resolution criterion; the proof supplies that criterion's full mathematical data. The general terminology and free quadratic presentation convention come from the cited definitions, not additional Lean axioms. In particular, [1, Theorem 8.5] is background only; its Poincare-series formula is not used as an assumed computation. Selected complete pages 24–27 of that survey and page 4 of Berger were inspected, not the complete papers.

Key declarations in namespace `TLMC561` are:

```
actualQuadraticPresentation
resolution
full_linear_resolution_certificate
graded_minimal_augmentation_certificate
ext_finrank
hilbert_identity
no_finite_cyclotomic_denominator
```

The exhaustive check enumerates the actual namespace, including generated declarations, and issues ordinary `#print axioms` for every name. Only `propext`, `Classical.choice` and `Quot.sound` are allowed; there is no `sorry`, `native_decide` or additional axiom.

The project pins Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`. From `lean/`, run `lean -version`, then `./verify.sh` for the build, warning-as-error axioms audit and official fresh-kernel replay. Equivalently run the latter three commands separately:

```
lake build +Main
lake env lean -DwarningAsError=true Check.lean
lake env leanchecker --fresh Main
```

The evidence directory records source-bound commands, logs and observed thread affinity; production timings are diagnostic, not a controlled research-efficiency result. Organizer acceptance is pending.

References

[1] J. McCullough and I. Peeva, *Infinite Graded Free Resolutions*, August 2018 survey, pp. 24–27, especially Definition 8.1 and the discussion of Yoneda algebras. https://pi.math.cornell.edu/~irena/papers/infiniteRes.pdf.

[2] R. Berger, *Gerasimov's theorem and N-Koszul algebras*, arXiv:0801.3383v3, 19 December 2008, p. 4, Definition 2.2. https://arxiv.org/abs/0801.3383.